Electro magnetic components upto my understanding

Iteration 1

1. electric current / magnetic fields :

When an electron is in its idle state an electric field is induced by it, when the electric charge is made to move, the magnetic field is generated by the electrons and the magnetic field will be in such a way that it depends on the flow of the direction of the electron. The electrons if flows in uni-direction which will be time invariant, then the generated magnetic field will be also time invariant meaning it will be a static magnetic field. But if the electrons are time variant i.e., if the electrons are moving in an sinusoidal direction , then the generated magnetic field will be also time variant and that means the generated magnetic field will be dynamic in nature.

2. Faradays and lenz law:

Faradays law states that when an changing magnetic field is introduced to a conductor material, an emf (electro motive force) will be induced in the conductor and the generated magnetic field will be opposing the source creating it (due to lenz law) i.e., the generated magnetic field in the copper conductor will be in the direction that is opposing the generated magnetic field. The emf equation is given by :

$$Emf = -N \frac{d\phi}{dt}.$$

This states that the magnetic field induced in the conductor will be in opposite direction (-ve sign) due to lenz law and will be based on the total number of conductor turns to the product of rate of change of applied magnetic flux with respect to time.

3. Amperes law :

Amperes law states that, when an current is passed to an conductor , an magnetic field will be induced in the conductor due to the flow of electrons. The integral equations means that in an closed loop of an conductor, one can achieve the total amount of current flowing in the circuit by integrating the field intensity of the close looped conductor material.

4. Inductor :

An inductor is an electromagnetic component. The main purpose of the inductor is to generate magnetic field and to store electric energy for a temporary period and discharge the electric current naturally as the magnetic field start to collapse. Mostly due to this characteristics the inductors are used as current source and filters. And used in power quality areas to make the power factor lagging to achieve appropriate power factor.

5. Inductance :

The inductance is the measure of how much the material can generate magnetic flux and store electrical energy in the generated magnetic field for an current which is applied to the terminals of it. The inductance is given by :

$L = N*\phi/I$ or L= flux linkage/ I

6. Flux linkage:

The flux linkage is the measure of how much magnetic field is hold by the conductors and it is found by multiplying the flux in the inductor to ne no.of conductor windings. The sum of total small conductor linkages ultimately form the total magnetic field generated by the inductor so if the number of turns is high, meaning the flux linkages will be also high so that the total magnetic field generated will also be high.

Flux linkage = number of conductor turns * flux

7. Flux density and field intensity :

Flux density is the total amount of flux lines generated by the inductor. The stronger the flux density the higher the inductance of the material, the unit is measured in T (Tesla). Field intensity (H) or magneto motive force is the intensity of the magnetic field. i.e., how fast the magnetic field is moving (Amperes/meter).

8. Energy stored in the inductor :

The energy stored in the inductor is the measure of how much joules of energy is stored in the core and the magnetic field of the inductor. W total = Wgap+Wcore , Wtotal = ½ * L* I^2. And it is measured in joules.

9. Effects of airgap and fringing effect:

Air is an low permeability medium meaning, to pass through the air medium the magnetic flux lines need a lot of push (more A/m) mmf. To cross the airgap the flux lines will try to connect to the core form all possible available directions which effectively increases the cross section area of the flux lines flowing in the leg where the air gap is present compared to the legs where air gap is not present and this effective area increased caused due to the airgap which increases the cross section area of the core virtually in the air medium is known as fringing effect.

The friging area can be calculated by :

Fringing = (a+g)(b+g)

A= cross section area of the core (width). B= cross section area of the core (depth) g= physical airgap length of the core.

The effect of airgap is that, it will virtually increase the saturation sealing of the material which means even though the core materials saturation point is low, by introducing an airgap , more current and high frequency can be passes to the core without saturating the core.

When, the conductors are placed at the leg where the airgap is present, the leakage flux and the fringing effect will be lower but if the windings are placed to the opposite legs, the fringing effect will be higher and so the leakage flux.

Need of core material :

- To generate high flux density with less conductors :

The main purpose of the core is that, it has to generate more magnetic field lines with less number of copper conductors than compared to air cores. By satisfying this property the total number of copper windings required to generate the flux will be lower meaning low i^2 r loss and compact size due to less copper turns and low stray capacitances between conductors .

- To offer low reluctance path (rm = l/mu0*mur*A)

The core has to offer low reluctance path meaning, the magnetic fields lines utilizes the total cross section area of the core as much as possible and due to this the area required by the core can be reduced. It the measure of how easy the core allows the flux lines to flow through it.

- To offer high resistance to eddy current

The core has to offer high resistance to eddy currents. This is an important factor to consider for high frequency applications as eddy current formation in the core increases if the frequency increases. So if the resistance offered to the eddy currents are low then the magnetic field lines due to skin effect will flow in the outer layer of the Core rather utilizing the cross section area fully which artificially saturates the core of the inductor fast as it virtually reduces the saturation point due to the skin effect and eddy current loops.

- To offer high permeability (r>>1)

the permeability is the measure of how easy an material can be magnetised. If the core has high permeability then the no of turns required to generate high inductance will be lower. If the material can get magnetised easily, then the current required will also be lower, which overall also reduces the size of the conductor. Vise versa occurs if the permeability is low .

factors considered for designing inductors :

- Saturation flux density (B)

  The saturation flux density is the measure of how much magnetic field the core can be able to generate as it approaches the point of its saturation. So cores are designed to be operated under the saturation point which is knee point it is the point where the core can generate maximum field without saturating the core. If the core material is energised even after its saturation point then the inductance will drop and the magnetic flux generated will start to collapse. Generally, the core is easy to magnetise from the period of not saturated to the point of being partially saturated because, at the initial when current is applied and magnetic flux lines are generated, the nanomagnetic particles in the core of the material will try to align with the generated magnetic field and at initial conditions the nanoparticles all point at random directions so it will be easy to align the particles to the applied magnetic fields with less current as the nanoparticles will not oppose the force acting on it. But from the point where the is partially saturated to fully saturated, at the point of partially saturated, the nanoparticles will align with the magnetic field applied partially and due to this they will start to oppose the magnetic field (same poles repel each other). So dur to this to fully align all the magnetic nano particles, more current is required to pass through the opposing force and thus it is hard to make all the particles align to the magnetic field. so to fully saturate the core, current

requirement will be high. So materials with high saturation points are preferred for most of the operations or if needed to delay the saturation, air gaps are induced. Materials like nanocrystalline based cores has high saturation point compared to other cores.

- Coercivity characteristics (Hc)

  The coercivity in the b-h curve means the current required to de magnetise the core material from its residual magnetism point. Usually when the current is removed, the core material will try to come to its original position of being an neutral material but due to the saturation process, the material will not become an neutral particle naturally as the nanoparticles inside the core will be still allighned to the path of the applied magnetic field. So residual magnetism will be present in the core. So , to bring the core back to its original state current is applied in the reverse direction in a way that is acting opposite to the previous applied current to make the core magnetise. So this applied current will force the nano materials in the core to alligh according to the applied field direction and thus at an particular point the core will come back to its original state. But the current has to be removed accurately or else the core will saturate in reverse direction. So, if lower coercivity means the loop of the hysteresis loss will be also lower which means low coercivity offers less hysteresis loss and vice versa. Material like amorphous has low hysteresis loss as it naturally has low coercivity characteristics but expensive compared to other cores

- Resistance characteristics

  The resistant characteristics directly affect the eddy current flow in the core of the material. If the resistance of the material is lower than the eddy current loops inside the core which will be formed automatically in the core due to self inductance and will be higher. And this eddy current loops will not allow the magnetic flux to flow in the cross section area of the core and forces it to flow through the outer layer of the core which virtually reduces the saturation point and the core saturates fast. The eddy current formation is also depends upon the frequency. If the frequency is high then the skin effect in the core and the eddy current will also be high and vice versa occurs, so for higher frequency high resistance materials like air or ferrite is used to the prevention of skin effect and eddy loops.

- Permeability characteristics

  Permeability is the measure of how easy the core can be magnetised. If the permeability is high then the material becomes magnet easily and vice versa occurs if the permeability is low. High permeability characteristics determine how much current and conductors turns required for the core. So permeability directly affects the materials nature of creating the magnetic field which is the inductance of the material. If permeability is high then no of copper conductor turns and the current required will be lower which ultimately reduces the size of the conductor.
  Core materials :
- Air core
- Solid iron core

- Laminated iron sheets core
- Amorphous core
- Nanocrystalline core
- Ferrite core
- Soft magnetic composite (SMC)

Air core :

- No saturation ceiling
- Low permeability and high reluctance
- High resistance to eddy current
- Stray capacitances only between the copper conductors
- No core so no core loss, no hysteresis loss
- Used for high frequency applications
- linear inductance characteristic

Solid iron core :

solid iron cores are not used as eddy currents will be so high at higher frequencies, skin effect occurs and also makes the magnetic flux lines to flow only through the outer layer and the core loss will be so high the core will be melted.

- Low saturation celing
- High permiablity and low reluctance
- Low resistance to eddy current
- Stray capacitances occur in between the core and copper as well as in the conductors loop.
- High core loss and hysteresis loss is high in higher frequencies
- Used for low frequency applications
- Non linear inductance but by introducing airgap, linear inductance can be achieved.

Laminated iron sheets core:

Since solid irons cannot be used, iron is chopped into thin sheets and then the thin sheets are coated with an layer of insulation and stacked together to form a single core.

2d magnetism characteristics can be observed in laminated iron sheet cores and it is purposefully done. In case of iron sheets, the material offers low resistance to the eddy currents. When the core is energised, the eddy currents will be formed in the cores due to self inductance but they cannot form a continuous large loop as the oxidated insulation layer will prevent the eddy currents and trap them only to their respective sheets, so eddy currents are drastically reduced also, this insulating layer allows the core to achieve 2d magnetism which means that the magnetic flux lines are forced by the insulating material to flow only in parallel paths and does not allow the magnetic lines to take the perpendicular direction, thus preventing 3d magnetism.

- resonable saturation celing
- High permiablity and low reluctance
- medium resistance to eddy current

- Stray capacitances occur in between the core and copper as well as in the conductors loop.
- reduced core loss and hysteresis loss due to insulation layer
- Used for medium frequency applications
- Non linear inductance but by introducing airgap, linear inductance can be achieved.
- Only 2D magnetism.

Amorphous core :

Very thin film layers which are cooled down immediately after removing from the furnace to prevent crystallization

- low coercivity, low hysteresis loss
- expensive compared to other cores
- high saturation ceiling
- high permeability and low reluctance
- reasonable resistance to eddy current
- core loss is less
- Used for medium frequency applications

Nanocrystalline cores :

- Lowest core loss
- Highest permeability and low reluctance
- Temperature celling is very low (150 deg c)
- High saturation celling
- High resistance to eddy current
- Used for medium frequency applications

Ferrite cores :

Materials like zinc and manganese are baked together to form a solid core

- Low core loss
- High permeability and low reluctance
- Mechanically and thermally fragile
- High saturation celling
- Highest resistance to eddy current
- Used for very high frequency applications
- Allows 3d magnetism safely

SMC (soft magnetic composite materials):

In this the high permeability material like iron is crushed and made to powder and insulation is added so that even the small nanoparticle of the core will be insulated and compressed together form the single sold core. The airgaps in the core is uniformly distributed .

- Low core loss
- reasonable permeability and mid reluctance (due to airgap and insulation)
- reasonable saturation celling

- High resistance to eddy current
- Used for medium frequency applications
- Allows 3d magnetism safely
- Fringing effect is reduced

3d magnetism is safely achieved in smc and ferrite cores. In the case of smc cores since the airgap is spread equally and each iron particle is coated uniformly the eddy currents are trapped in the airgaps and they cannot form a strong loop as the insulating material prevents the eddy formation. So magnetic flux lines can flow in the both parallel and perpendicular directions. In case of ferrite cores, the core is made up of non conducting material so, 3d magnetism can cause no effect on the core and it naturally offers high eddy current.

Transformers :

The primary difference between the choke and the transformer is that the choke strores and releases electric current as its magnetic fields forms and generates respectively whereas the transformer does not store any energy in the magnetic field and only transforms it by stepping it up or down.

Ideal transformers :

An transformer is said to be ideal if there is no core loss, copper loss, leakage flux, inrush current and only pure sinusoidal waveform is taken into account.

When the primary of the ideal transformer is connected to the source at no load, the pure sinusoidal primary voltage will flow through the transformer and this voltage will create an current component known as magnetising current in the primary side of the transformer. This magnetising component will be 90 degree lagging to the applied voltage u1 as the transformer is a basically acting as an lagging component. This lagging magnetising current component will generate an alternating flux in the core which will be inphase with the magnetising current. Since it is an alternating magnetic flux and due to faradays law the alternating flux creates an electro motive force in the core which will be opposing the primary voltage u1 and will be 90 degree lagging to the magnetising current and alternating flux. This emf is known as back emf of the transformer. The magnitude of the back emf and the applied u1 will be same.

Core loss :

The magnetising current i0 will be the sum of magnetising component plus the iron loss component. The iron loss component supplies to the core loss i.e., hysteris and eddy loss that is occurring in the core and ultimately leading to heat loss. Since heat loss is generated by real power the iw will be in phase with the applied voltage u1., the magnetising component supplies to the generation of magnetic flux inside the core. So, transformer naturally draws the current in the primary side such that it satisfies both the components.

Copper loss and lekage flux :

In practical applications, copper loss occurs and lekage flux occurs, the copper loss is caused due to the internal copper resistances and w.k.t not all flux generated will be useful for work, some may leak so it is represented as lekage flux. The voltage drop caused by the copper losses and voltage drop caused by the lekage flux are taken to account. The lekage flux will be leaked in air and this leads to be in phase with the magnetising current component. the volage has to flow

through the copper to reach core so the voltage in the terminals will be higher and the voltage in the core will be low due to these voltage drops.

1 phase transformer at loaded conditions :

When an load is connected to the secondary side of the transformer there will be an emf and current that will be induced in the secondary and it will be based on transformer ratio and the total number of copper turns. When primary is supplied, now the primary current has to satisfy the magnetising current as well as the secondary current as the secondary current will induce an alternating flux and in turn it induces and back emf in the secondary which will oppose the back emf in primary thus to balance the emfs, the transformer automatically draws extra current. So primary current is the sum of secondary current + magnetising component current.

The primary side applied voltage at the terminals up will be higher and the primary core voltage um1 will be lower as it has to pass through the voltage drops of resistance and reactance. Also, in the secondary side core voltage um2 will be high as it is generated there but when the terminal voltage is to be achieved, it has to flow through the internal resistance and lekage flux so the terminal voltge u2 will be lower.

N1 = N2 : V and I on the primary and secondary will be same

N1>N2 (step down): in primary V will be high and I will be low and in secondary : V will be low and I will be high

N1<N2 (step up): in primary V will be low and I will be high and in secondary : V will be high and I will be low.

I12= I2/k(transformer ratio).

Transformer inrush current :

In ideal conditions When the primary of the ideal transformer is connected to the source at no load, the pure sinusoidal primary voltage will flow through the transformer and this voltage will create an current component known as magnetising current in the primary side of the transformer. This magnetising component will be 90 degree lagging to the applied voltage u1 as the transformer is a basically acting as an lagging component. This lagging magnetising current component will generate an alternating flux in the core which will be inphase with the magnetising current. Since it is an alternating magnetic flux and due to faradays law the alternating flux creates an electro motive force in the core which will be opposing the primary voltage u1 and will be 90 degree lagging to the magnetising current and alternating flux. This emf is known as back emf of the transformer. The magnitude of the back emf and the applied u1 will be same. So, the primary voltage and magnetic field (phi) will be symmetrical. And in ideal conditions the phi waveform starts from -ve peak. The emf equation according to faraday is given by :

U1= n*dphi/ dt and Bemf = -n*dphi/dt.

In real life conditions :

When the transformer is connected to the primary, the voltage of primary can be at any point in the sinusoidal waveform. But when the magnetic field is started to generate, it either has to start from 0 or start from the point of residual magnetism. So to satisfy the faradays law as well as the

boundary conditions, the magnetic flux starts to behave in sinusoidal pattern which induces dc offset voltage in the flux wave and its value will be the instantaneous value of the input voltage at initial condition.

So, this makes the peak flux to be increased to 2 times if it started from 0 or 2 times peak + residual flux if started from the residual magnetism point. Due to this the peak of the flux will be very high than the core is made to withstand and because of this the core will saturate and will exceed the saturation point and so that the magnetic field and inductance inside the core start to collapse and this leads to further consumption of magnetising current in the primary side and this transient current that is drawn to tackle the collapse of the inductance in the core after the saturation point is crossed is known as the inrush current. This phenomenon occurs only at the starting period of the transformer.

Effects :

If inrush current is not taken into account while desighning protective devices for the transformer, tripping occurs commonly which will affect the reliability of the grid.

Prevention:

In 1 phase transformer, when the primary is connected to the voltage when the voltage is at +ve peak or -ve peak, the inrush magnetising current is completely prevented.

In 3 ph transformer inrush current always occurs and cannot be prevented as each phase is 120 degree apart.